Aligned Mathematical Structures

Source-backed grouped formula sample

Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$F^\dagger H^\dagger H F(p,q) = \delta(p,q)\begin{pmatrix} 3 & 0 & 2e^{-ip_1+ip_2} & 2e^{-ip_1-ip_2} \\ 0 & 3 & -2e^{+ip_1+ip_2} & 2e^{+ip_1-ip_2} \\ 2e^{+ip_1-ip_2} & -2e^{-ip_1-ip_2} & 3 & 0 \\ 2e^{+ip_1+ip_2} & 2e^{-ip_1+ip_2} & 0 & 3 \end{pmatrix},$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{I}(\lambda\delta + (1-\lambda)\delta') &= \sup_{\nu \in \Sigma(\lambda\delta+(1-\lambda)\delta')} \int_{\mathcal{S}} f(s)d\nu(s) \\ &\geq \int_{\mathcal{S}} f d\gamma'' = \lambda \int_{\mathcal{S}} f d\gamma + (1-\lambda)\int_{\mathcal{S}} f d\gamma'. \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\frac{1}{2}\left\{ \begin{bmatrix} Q_\alpha \\ \tilde{Q}_\alpha \end{bmatrix}, \begin{bmatrix} Q_\beta & \tilde{Q}_\beta \end{bmatrix} \right\} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} M\delta_{\alpha\beta} + \begin{bmatrix} p & q/g_s \\ q/g_s & -p \end{bmatrix} \frac{L(\Gamma^0\Gamma^1)_{\alpha\beta}}{2\pi\alpha'}.$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \delta\Psi_\alpha &= \frac{1}{8}\omega^{mn}(\Gamma_{mn})_\alpha^{\ \beta}\Psi_\beta, \quad \Gamma_{mn} \equiv \Gamma_m\tilde{\Gamma}_n - \Gamma_n\tilde{\Gamma}_m, \\ \delta\chi^\alpha &= \frac{1}{8}\omega^{mn}(\tilde{\Gamma}_{mn})^\alpha_{\ \beta}\chi^\beta, \quad \tilde{\Gamma}_{mn} \equiv \tilde{\Gamma}_m\Gamma_n - \tilde{\Gamma}_n\Gamma_m, \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$R_{ij}^{(\ell)}(\Lambda_1, ..., \Lambda_i, ..., \Lambda_j, ..., \Lambda_N) = \begin{cases} (\Lambda_1, ..., \Lambda_i - \ell, ..., \Lambda_j + \ell, ..., \Lambda_N) \text{ if } & \Lambda_i > \Lambda_j, \\ (\Lambda_1, ..., \Lambda_i + \ell, ..., \Lambda_j - \ell, ..., \Lambda_N) \text{ if } & \Lambda_j > \Lambda_i. \end{cases}$$